# PCA and Dimensionality Reduction

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## Part I — Free-Response Questions (FRQs): Foundational Fluency (1–7)

**1** [Foundational] • Target: PCA & Dimensionality Reduction Practice Exercises

A health study surveys 5,000 adults and finds that people who sleep 8 hours per night have significantly lower blood pressure than those who sleep 5 hours. (a) Can researchers conclude that increasing sleep causes a reduction in blood pressure? (b) Delineate the scope of inference using the 2×2 selection/assignment matrix.

**2** [Foundational] • Target: PCA & Dimensionality Reduction Practice Exercises

Level 3 · Advanced Find the full interval of convergence (including endpoint behavior) for the power series $\sum_{n=1}^{\infty} \frac{(2x-1)^n}{n^2 \cdot 4^n}$.

**3** [Foundational] • Target: PCA & Dimensionality Reduction Practice Exercises

Level 2 · MediumDetermine whether the polynomials $p_1(x) = 1 + x$, $p_2(x) = x + x^2$, and $p_3(x) = 1 + x^2$ form a basis for $\mathcal{P}_2$ (the vector space of polynomials of degree at most 2).

**4** [Foundational] • Target: PCA & Dimensionality Reduction Practice Exercises

Let $T : \mathbb{R}^2 \to \mathbb{R}^2$ have standard matrix $A = (4, 1; 2, 3)$. Find the matrix representation $[T]_{\mathcal{B}}$ relative to the basis $\mathcal{B} = \{(1; 1), (-1; 2)\}$ via similarity transformation $P^{-1}AP$.

**5** [Foundational] • Target: PCA & Dimensionality Reduction Practice Exercises

An agronomist wants to test whether a new organic fertilizer increases tomato yield compared to a standard commercial fertilizer. Explain how the agronomist must incorporate the four principles of experimental design: Comparison, Random Assignment, Control, and Replication.

**6** [Foundational] • Target: PCA & Dimensionality Reduction Practice Exercises

Let $\mathbf{u} = (1, -2, 2, 0)^T$ and $\mathbf{v} = (2, 1, 0, -2)^T$ in $\mathbb{R}^4$. Compute $||\mathbf{u}||$, $||\mathbf{v}||$, the inner product $\mathbf{u} \cdot \mathbf{v}$, the cosine of the angle $\theta$ between them, and verify the Cauchy–Schwarz inequality.

**7** [Foundational] • Target: PCA & Dimensionality Reduction Practice Exercises

Level 3 · Advanced A particle moves along a curve defined parametrically by $x = t^2 + 1$ and $y = t^3 - 3t$. Find the value of $\frac{d^2y}{dx^2}$ at $t = 2$ and determine whether the curve is concave up or concave down at this point.

## Part II — Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)

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**8** [Intermediate] • Target: PCA & Dimensionality Reduction Practice Exercises

A school district with 12 high schools wants to sample 300 teachers. Compare: (a) Stratified sampling (taking an SRS of 25 teachers from each of the 12 schools) vs. (b) Cluster sampling (randomly selecting 2 schools and surveying all 150 teachers in each). Discuss sampling variability and logistical practicality.

**9** [Intermediate] • Target: PCA & Dimensionality Reduction Practice Exercises

Evaluate the integral $\int_0^1 \int_y^1 e^{x^2}\,dx\,dy$ by reversing the order of integration.

**10** [Intermediate] • Target: PCA & Dimensionality Reduction Practice Exercises

Level 2 · MediumFind the minimum number of subintervals $n$ required to guarantee that the Trapezoidal approximation of $\int_1^3 \frac{1}{x}\,dx$ has an absolute error less than $10^{-4}$.

**11** [Intermediate] • Target: PCA & Dimensionality Reduction Practice Exercises

Level 2 · Medium Find the total arc length of the parametric curve $x = \cos^3 t$, $y = \sin^3 t$ (the astroid) for $0 \le t \le \frac{\pi}{2}$.

**12** [Intermediate] • Target: PCA & Dimensionality Reduction Practice Exercises

Level 3 · AdvancedA university alumni association randomly selects 500 graduates from its directory and sends them a survey asking for their current annual salary. Only 95 completed surveys are returned, yielding an average reported salary of 115,000. (a) Name the primary source of bias present. (b) Explain the likely direction of the bias on the estimated mean salary. (c) Explain whether the results can be generalized to all graduates of the university.

**13** [Intermediate] • Target: PCA & Dimensionality Reduction Practice Exercises

Level 1 · FoundationsVerify that the Mean Value Theorem applies to $f(x) = x + \frac{1}{x}$ on $[1, 4]$, and find all values of $c$ that satisfy the theorem.

**14** [Intermediate] • Target: PCA & Dimensionality Reduction Practice Exercises

Find the least-squares solution $\hat{\mathbf{x}}$ to the inconsistent system $A\mathbf{x} = \mathbf{b}$, where

$A = (1, 1; 1, 2; 1, 3), \qquad \mathbf{b} = (1; 2; 4).$

## Part III — Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20)

**15** [Challenging] • Target: PCA & Dimensionality Reduction Practice Exercises • Cross-Lesson: 1.01 Vectors and Embeddings in AI

Level 3 · Advanced Use the Divergence Theorem to find the outward flux of $\mathbf{F} = \langle x^3, y^3, z^3 \rangle$ across the boundary sphere $S : x^2 + y^2 + z^2 = 1$.

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**16** [Challenging] • Target: PCA & Dimensionality Reduction Practice Exercises • Cross-Lesson: 2.01 Neural Layer Matrices and Activations

The commute times for employees at a large tech company have a strongly right-skewed distribution with mean $\mu = 38$ minutes and standard deviation $\sigma = 15$ minutes. A random sample of 49 employees is selected. (a) Describe the shape, mean, and standard error of the sampling distribution of the sample mean $\bar{x}$. (b) Calculate $P(\bar{x} > 42 \text{ minutes})$.

**17** [Challenging] • Target: PCA & Dimensionality Reduction Practice Exercises • Cross-Lesson: 3.01 Gradient Descent and Optimization

Level 2 · MediumEvaluate the trigonometric limit: $\lim_{x \to 0} \frac{1 - \cos(3x)}{x^2}$.

**18** [Challenging] • Target: PCA & Dimensionality Reduction Practice Exercises • Cross-Lesson: 4.01 Conditional Probability and Bayes Rule

Level 3 · Advanced Find the exact surface area generated by revolving the line segment $y = 2x$ on $[0, 3]$ about the $x$-axis.

**19** [Challenging] • Target: PCA & Dimensionality Reduction Practice Exercises • Cross-Lesson: 5.01 Distributions and Maximum Likelihood

Level 2 · MediumSolve the transcendental equation analytically for $x$ in exact form using natural logarithms: $3^{2x+1} = 5^{x-2}$.

**20** [Challenging] • Target: PCA & Dimensionality Reduction Practice Exercises • Cross-Lesson: 6.01 Linear Regression and OLS

Use Green's Theorem to evaluate $\oint_C (y^2 - 7y)\,dx + (2xy + 2x)\,dy$, where $C$ is the positively oriented boundary of the region $D$ enclosed between the parabolas $y = x^2$ and $x = y^2$.

## Part IV — Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

**21** [Foundational] • Target: PCA & Dimensionality Reduction Practice Exercises

A pharmaceutical company is testing a new drug to reduce blood pressure. They conduct a clinical trial with $n = 100$ participants. The sample mean reduction in systolic blood pressure is $\bar{x} = 15$ mmHg with a sample standard deviation of $s = 8$ mmHg. The company plans to construct a 95% confidence interval for the true mean reduction in systolic blood pressure. If the margin of error for this confidence interval is $M = 2$ mmHg, what would be the required sample size, $n'$, to achieve a margin of error of $M' = 1$ mmHg with the same confidence level and an estimated population standard deviation equal to the sample standard deviation?

**a.** 200

**b.** 300

**c.** 400

**d.** 1600

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**22** [Foundational] • Target: PCA & Dimensionality Reduction Practice Exercises

A pharmaceutical company is conducting a clinical trial to test the efficacy of a new drug designed to lower blood pressure. They recruit 200 participants and randomly assign them to either the treatment group (receiving the new drug) or the control group (receiving a placebo). After 12 weeks, they measure the reduction in systolic blood pressure for each participant. The company wants to perform a hypothesis test to determine if there is a significant difference in blood pressure reduction between the two groups. The trial is designed with 20 participants in the treatment group and 180 participants in the control group. If the standard deviation of blood pressure reduction in the treatment group is $\sigma_1 = 15$ mmHg and in the control group is $\sigma_2 = 12$ mmHg, and they are planning to use a two-sample t-test, what is the calculated pooled standard deviation $s_p$ if equal variances are assumed, given the following summary statistics for blood pressure reduction (in mmHg): $\bar{x}_1 = 10$ mmHg for the treatment group and $\bar{x}_2 = 7$ mmHg for the control group?

**a.** $\sqrt{\frac{15^2 + 12^2}{2}}$

**b.** $\sqrt{\frac{20(15^2) + 180(12^2)}{200}}$

**c.** $\frac{20 \times 15 + 180 \times 12}{200}$

**d.** $\sqrt{\frac{19(15^2) + 179(12^2)}{198}}$

**23** [Foundational] • Target: PCA & Dimensionality Reduction Practice Exercises

Reduction formula: $\int \sin^n x\,dx = -\frac{\sin^{n-1} x \cos x}{n} + \frac{n-1}{n}\int \sin^{n-2} x\,dx$. Then $\int_0^{\pi/2} \sin^6 x\,dx =$

**a.** $5\pi/16$

**b.** $3\pi/16$

**c.** $5\pi/32$

**d.** $15\pi/48$

**24** [Foundational] • Target: PCA & Dimensionality Reduction Practice Exercises

Using the reduction formula, evaluate $\int_0^{\frac{\pi}{2}} \cos^6(x)\,dx$.

**a.** $\frac{5\pi}{16}$

**b.** $\frac{5\pi}{32}$

**c.** $\frac{3\pi}{16}$

**d.** $\frac{1}{6}$

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**25** [Foundational] • Target: PCA & Dimensionality Reduction Practice Exercises

Reduction formula: $I_n = \int \sin^n x\,dx$ satisfies:

**a.** $I_n = -\frac{\sin^{n-1} x \cos x}{n} + \frac{n-1}{n} I_{n-2}$

**b.** $I_n = \frac{\sin^{n+1} x}{n+1} + C$

**c.** $I_n = -n \cos x \sin^{n-1} x + C$

**d.** $I_n = I_{n-1} \cdot I_1$

**26** [Foundational] • Target: PCA & Dimensionality Reduction Practice Exercises

Reduction formula for $\int \sin^n x\,dx$ comes from IBP with $u =$

**a.** $\sin^n x$

**b.** $\cos x$

**c.** $\sin^{n-1} x$

**d.** $x$

**27** [Foundational] • Target: PCA & Dimensionality Reduction Practice Exercises

Reduction of order: given $y_1 = e^x$ solves $y'' - y = 0$. Second independent solution via $y_2 = v(x)e^x$:

**a.** $y_2 = xe^x$

**b.** $y_2 = e^x \ln x$

**c.** $y_2 = xe^{-x}$

**d.** $y_2 = e^{-x}$

**28** [Foundational] • Target: PCA & Dimensionality Reduction Practice Exercises

$\int \sin^4 x\,dx$ using reduction formula:

**a.** $\frac{3x}{4} + \frac{\sin 2x}{4} + C$

**b.** $\frac{3x}{8} - \frac{\sin 2x}{4} + \frac{\sin 4x}{32} + C$

**c.** $\frac{x}{2} - \frac{\sin 4x}{8} + C$

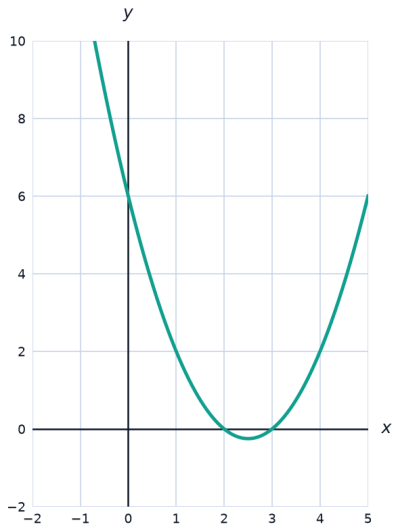
**d.** $\frac{\sin^3 x \cos x}{4} + \frac{\sin x \cos x}{4} + \frac{x}{8} + C$

## Part V — Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

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**29** [Intermediate] • Target: PCA & Dimensionality Reduction Practice Exercises

Consider the function $f(x) = x^2 - 5x + 6$. What is the derivative of $f(x)$ with respect to $x$?




**a.** $2x - 5$

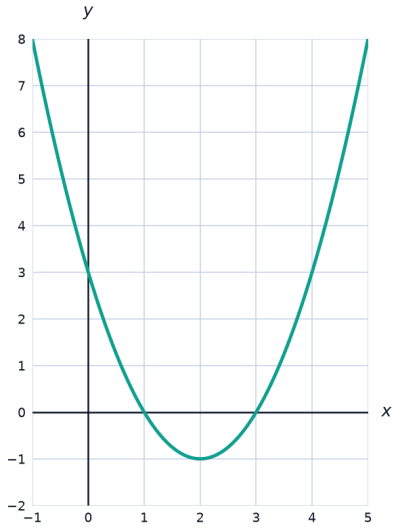
**b.** $x - 5$

**c.** $x^2 - 5$

**d.** $2x$

**30** [Intermediate] • Target: PCA & Dimensionality Reduction Practice Exercises

The graph of the function $f(x) = x^2 - 4x + 3$ is shown below. The function has a local minimum. What are the coordinates of the local minimum?




**a.** (2, 1)

**b.** (2, -1)

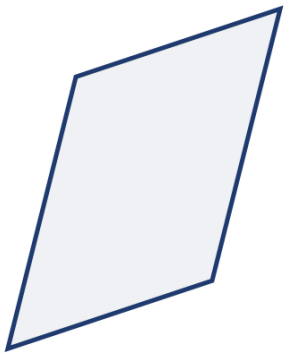
**c.** (1, 0)

**d.** (3, 0)

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**31** [Intermediate] • Target: PCA & Dimensionality Reduction Practice Exercises

A parallelogram in the xy-plane is defined by the vectors $\mathbf{u} = (3; 1)$ and $\mathbf{v} = (1; 4)$. What is the area of this parallelogram?



**a.** 10

**b.** 11

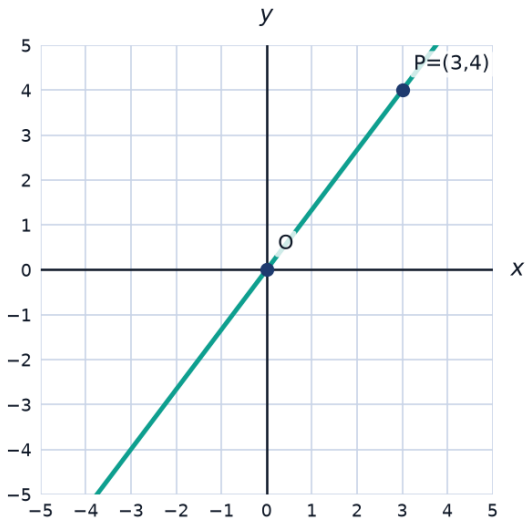
**c.** 13

**d.** 14

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**32** [Intermediate] • Target: PCA & Dimensionality Reduction Practice Exercises

Consider the vector $\mathbf{v} = (3; 4)$. If this vector is plotted on a standard $xy$-plane, it represents a directed line segment from the origin to the point $(3, 4)$. What is the geometric interpretation of the length of this vector?




**a.** The vector's direction in the xy-plane.

**b.** The angle the vector makes with the positive x-axis.

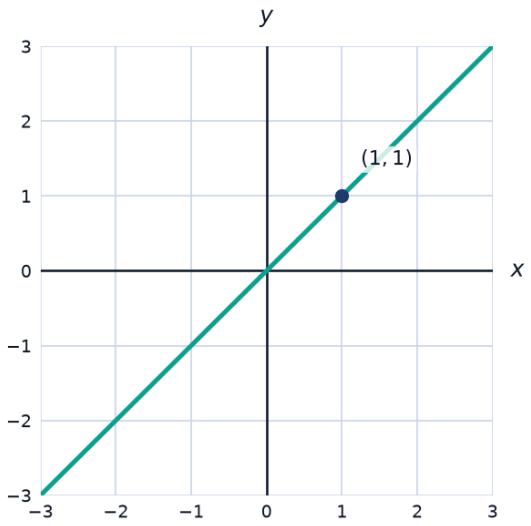
**c.** The distance of the point (3,4) from the origin.

**d.** The sum of the components of the vector.

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**33** [Intermediate] • Target: PCA & Dimensionality Reduction Practice Exercises

Consider the graph of the function $f(x)$ shown below. What is the limit of $f(x)$ as $x$ approaches 1?




**a.** 1

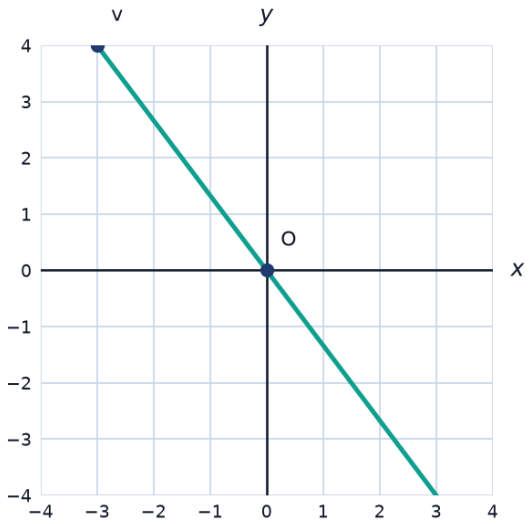
**b.** 0

**c.** 2

**d.** Does not exist

**34** [Intermediate] • Target: PCA & Dimensionality Reduction Practice Exercises

Consider the vector $\mathbf{v} = (-3; 4)$ in $\mathbb{R}^2$. What is the magnitude (or norm) of this vector?




**a.** 5

**b.** 7

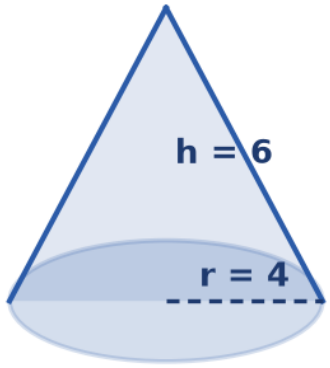
**c.** 1

**d.** 25

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**35** **[Intermediate]** • **Target: PCA & Dimensionality Reduction Practice Exercises**

Water pours into an inverted cone (radius 4 m, height 6 m) at 4 m³/min. How fast is the depth rising when it is 4 m? (m/min)




**a.** $\frac{3}{8\pi}$

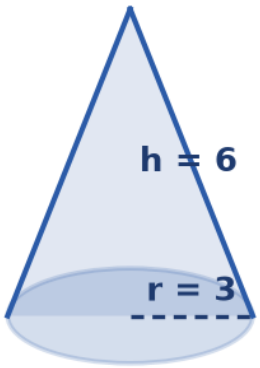
**b.** $\frac{9}{4\pi}$

**c.** $\frac{9}{16\pi}$

**d.** $\frac{1}{4\pi}$

**36** **[Intermediate]** • **Target: PCA & Dimensionality Reduction Practice Exercises**

Water drains from a cone-shaped tank (radius $3$, height $6$) at $\frac{dV}{dt} = 2$ m$^3$/min. Using $V = \frac{1}{3}\pi r^2 h$ with $r = \frac{3}{6}h$, find $\frac{dh}{dt}$ when $h = 2$.




**a.** $\frac{2}{\pi}$

**b.** $\frac{1}{\pi}$

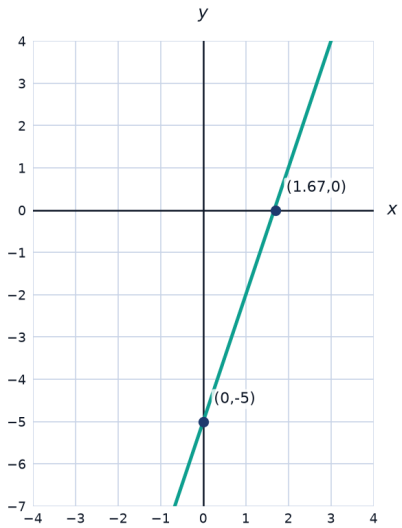
**c.** $-\frac{2}{\pi}$

**d.** $\frac{4}{\pi}$

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**37** **[Intermediate]** • **Target: PCA & Dimensionality Reduction Practice Exercises**

The slope of $y = 3x - 5$ at any point is:




**a.** $-5$

**b.** 0

**c.** 3

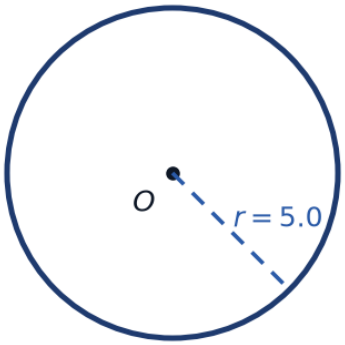
**d.** undefined

## Part VI — Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45)

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**38** [Challenging] • Target: PCA & Dimensionality Reduction Practice Exercises • Cross-Lesson: 4.01 Conditional Probability and Bayes Rule

Circle centred at origin, radius 5:




**a.** $x^2 + y^2 = 5$

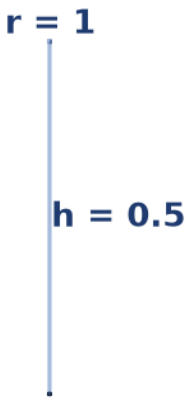
**b.** $x^2 + y^2 = 25$

**c.** $(x - 5)^2 + y^2 = 0$

**d.** $y = \pm\sqrt{25 - x^2}$

**39** [Challenging] • Target: PCA & Dimensionality Reduction Practice Exercises • Cross-Lesson: 5.01 Distributions and Maximum Likelihood

A water tank (cylinder, radius 1 m) drains. If height decreases at 0.5 m/min, water volume leaving per minute:




**a.** $0.5\pi$

**b.** $\pi$

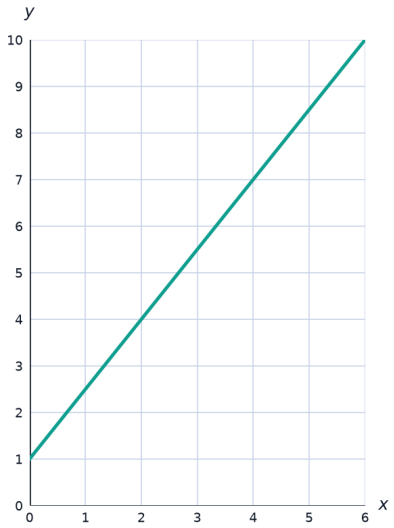
**c.** $2\pi$

**d.** $4\pi$

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**40** [Challenging] • Target: PCA & Dimensionality Reduction Practice Exercises • Cross-Lesson: 6.01 Linear Regression and OLS

In $\hat{y} = a + bx$, the slope $b$ represents:




**a.** The predicted value at x=0

**b.** Change in y per unit x

**c.** Correlation coefficient

**d.** Standard error

**41** [Challenging] • Target: PCA & Dimensionality Reduction Practice Exercises • Cross-Lesson: 7.01 Logistic Regression and Classification

Consider a particle whose velocity vector $\mathbf{v}(t) = \langle x'(t), y'(t) \rangle$ satisfies the differential equation $y'' - 2y' + y = 0$ and the initial conditions $x(0) = 1, y(0) = 2, x'(0) = 0, y'(0) = 1$. Furthermore, the particle's path is such that the magnitude of its acceleration vector is always equal to the magnitude of its velocity vector, i.e., $||\mathbf{a}(t)|| = ||\mathbf{v}(t)||$. Find the value of $x(1)$.

**a.** $e^{-1} + 1$

**b.** $2e^{-1} + 1$

**c.** $e^{-1} - 1$

**d.** $2 - e^{-1}$

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**42** [Challenging] • Target: PCA & Dimensionality Reduction Practice Exercises • Cross-Lesson: 1.01 Vectors and Embeddings in AI

A team of statisticians is designing a complex experiment to evaluate the efficacy of a new fertilizer on crop yield across multiple regions and soil types. They have identified 3 distinct geographic regions and 5 different soil types. Within each region-soil type combination, they plan to test 4 different concentrations of the fertilizer. To account for variability, they will run 2 independent replicates of each treatment combination. However, due to resource limitations, they can only afford to run 150 unique treatment combinations in total. The team decides to allocate the available resources by prioritizing the combinations that are most likely to reveal significant differences, while ensuring a balanced representation across the major factors. They will select a subset of the possible treatment combinations. If they decide to use a stratified sampling approach where each region is a stratum, and within each region, they stratify by soil type, and then sample fertilizer concentrations, what is the minimum number of unique treatment combinations (region, soil type, concentration) they *must* include in their sample to ensure that every soil type is represented in at least one region's stratum, and every fertilizer concentration is represented within each soil type stratum across all regions, given that they still wish to have 2 replicates per selected combination?

**a.** 150

**b.** 90

**c.** 75

**d.** 60

**43** [Challenging] • Target: PCA & Dimensionality Reduction Practice Exercises • Cross-Lesson: 2.01 Neural Layer Matrices and Activations

Consider the sequence defined by $a_1 = 1$ and $a_{n+1} = \frac{a_n^2 + k}{2}$ for $n \geq 1$, where $k$ is a positive constant. If the sequence converges to a limit $L$, prove that $L^2 - 2L + k = 0$. Furthermore, if the limit $L$ is an integer and $k = 3$, determine the exact value of the limit.

**a.** L = 3

**b.** L = 1

**c.** L = -1

**d.** $L = \sqrt{3}$

**44** [Challenging] • Target: PCA & Dimensionality Reduction Practice Exercises • Cross-Lesson: 3.01 Gradient Descent and Optimization

Consider the vector field $\mathbf{F}(x, y) = \langle 2x, y \rangle$. Evaluate the line integral $\oint_C \mathbf{F} \cdot d\mathbf{r}$ where $C$ is the circle $x^2 + y^2 = 4$ oriented counterclockwise. Use Green's Theorem.

**a.** 0

**b.** $8\pi$

**c.** $4\pi$

**d.** $16\pi$

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**45** [Challenging] • Target: PCA & Dimensionality Reduction Practice Exercises • Cross-Lesson: 4.01 Conditional Probability and Bayes Rule

Consider the matrix $A = (k, 2; 1, -3)$. If the matrix $A$ is invertible and its inverse $A^{-1}$ has a determinant of $-\frac{1}{11}$, find the value of $k$.

**a.** 4

**b.** 5

**c.** 3

**d.** 6

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